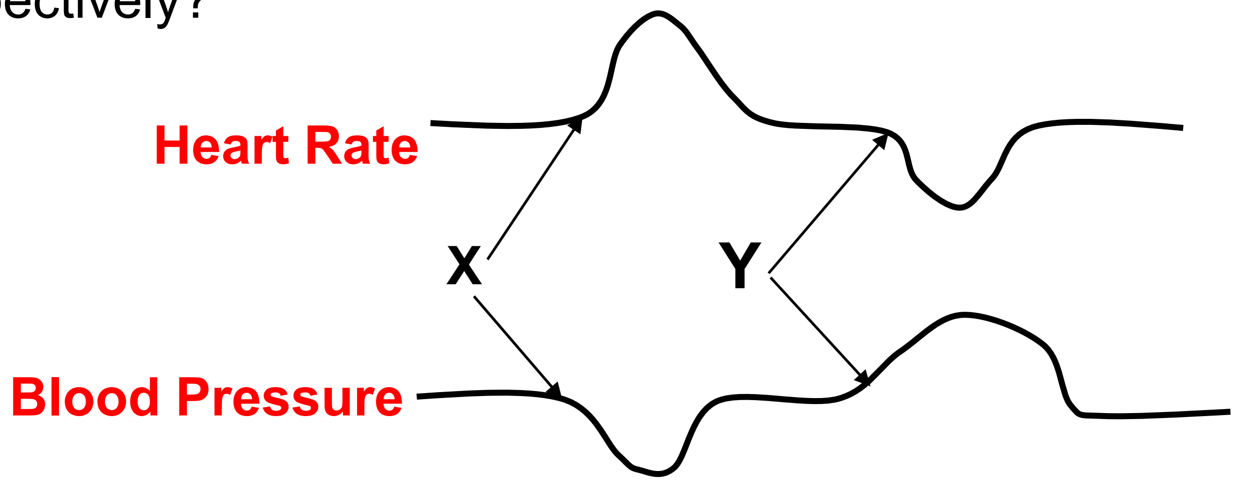
# FTM IMCQ 9
# Fall 2025

KEY File

# FTM IMCQ 9

4 PHARM Questions

A team of pharmacologists is studying the effects of drug X and drug Y on blood pressure and heart rate in rats. Drug X is administered, and then, after its effects have disappeared, drug Y is administered. The figure below shows the changes observed in mean arterial pressure (MAP) and heart rate. Which of the following are most likely drug X and drug Y, respectively?




A. Acetylcholine and phentolamine

B. Norepinephrine and bethanechol

C. Norepinephrine and phenylephrine

D. Bethanechol and phenylephrine

E. Epinephrine high dose and norepinephrine

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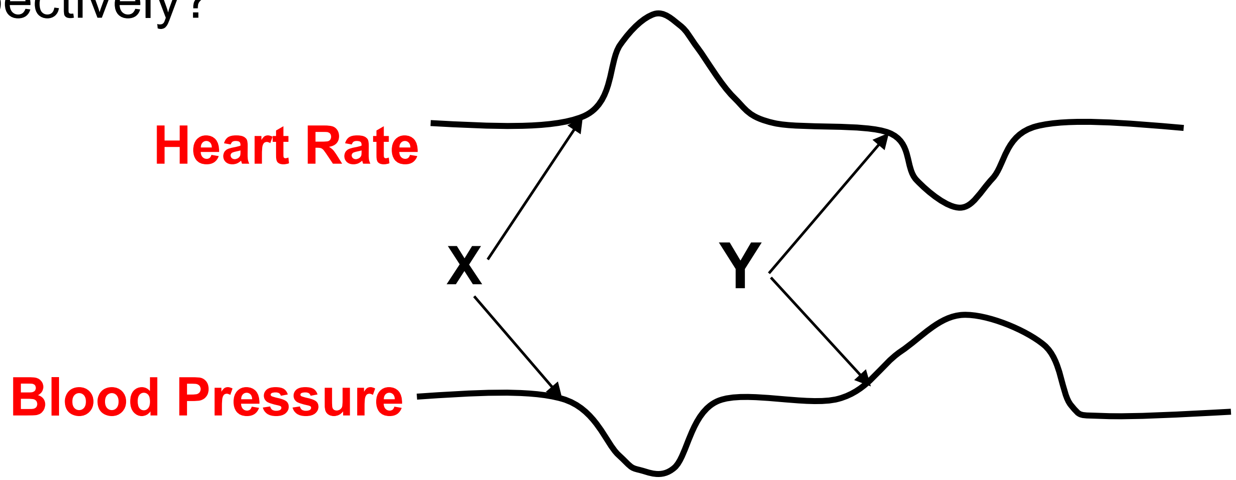


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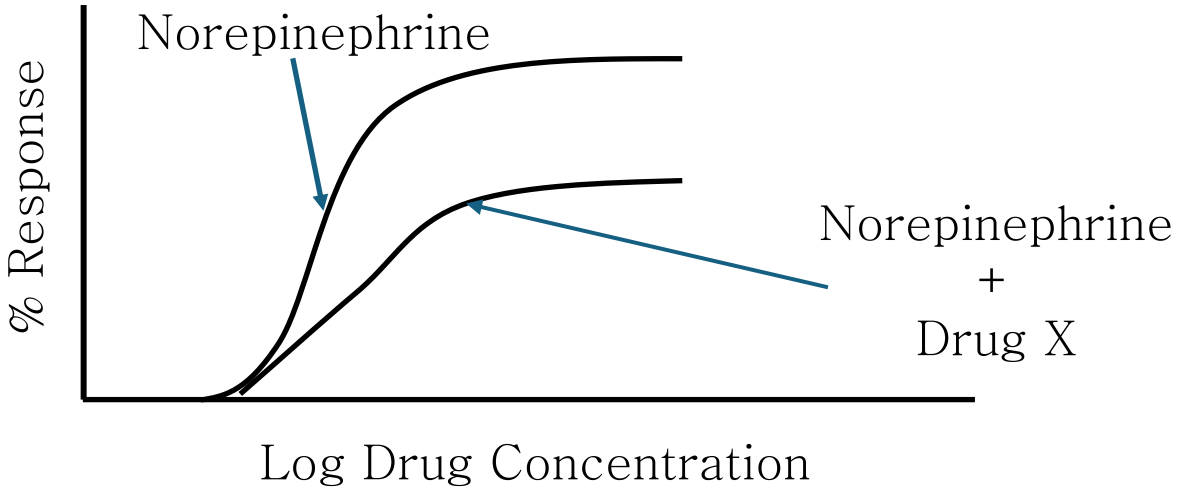
SOM.MK.1.BPM1.1.FTM.4.PHAR.0056 Describe the pharmacological effects, uses and adverse effects of alpha1 and alpha2 agonists.

SOM.MK.1.BPM1.1.FTM.4.PHAR.0060Describe the pharmacological effects, uses and adverse effects ofbeta adrenergic antagonists. Compare and contrast the adverseeffects of the nonspecific and cardiospecific beta adrenergic antagonists.

A team of pharmacologists is studying the effects of a novel drug, Drug X, on alpha adrenergic receptors. The concentration response curves of norepinephrine and Drug X plus norepinephrine are shown in the diagram below.  Which of the following drugs is most likely similar to drug X?




A. Epinephrine high dose

B. Albuterol

C. Phentolamine

D. Phenoxybenzamine

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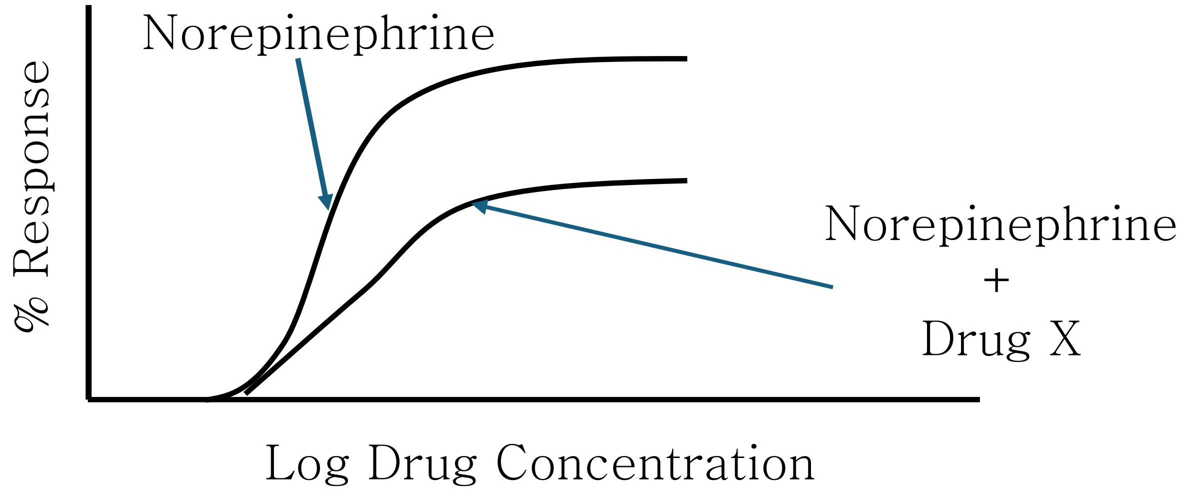


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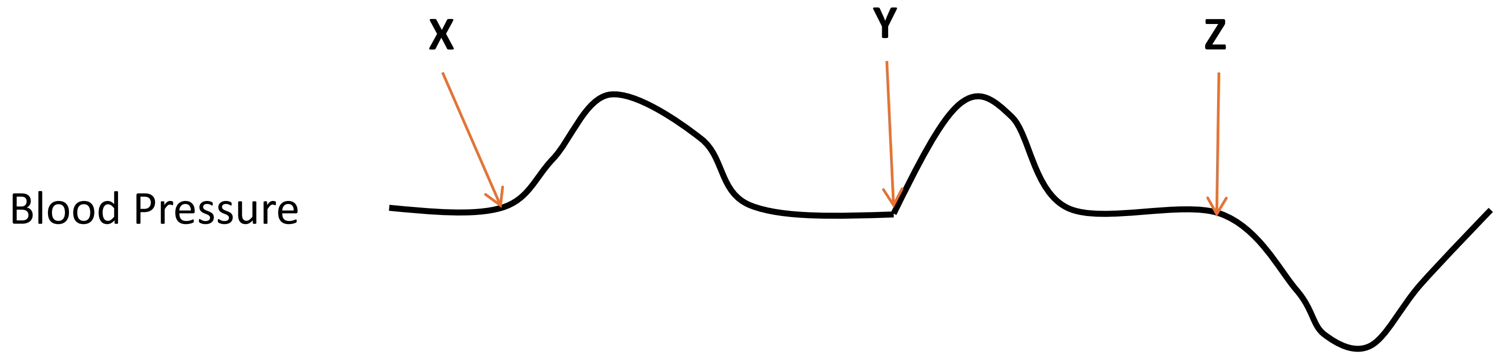
SOM.MK.1.BPM1.1.FTM.4.PHAR.0056 Describe the pharmacological effects, uses and adverse effects of alpha1 and alpha2 agonists.

A team of pharmacologists is studying the effects of Drugs X, Y, and Z on blood pressure in rats. First, Drug X is administered, and after its effect subsides Drug Y is administered. When the effect of Drug Y subsides, Drug Z is administered. The graph below shows the changes in blood pressure. The arrows show the times when the drugs were administered. Which of the following sets of drugs listed in the options below is most likely to correspond to Drugs X, Y, and Z?




| | X | Y | Z |
|---|---|---|---|
| A. | Norepinephrine | Epinephrine (Low Dose) | Acetylcholine |
| B. | Phenylephrine | Acetylcholine | Prazosin |
| C. | Prazosin | Norepinephrine | Acetylcholine |
| D. | Norepinephrine | Phenylephrine | Phentolamine |
| E. | Acetylcholine | Prazosin | Epinephrine (Low dose) |

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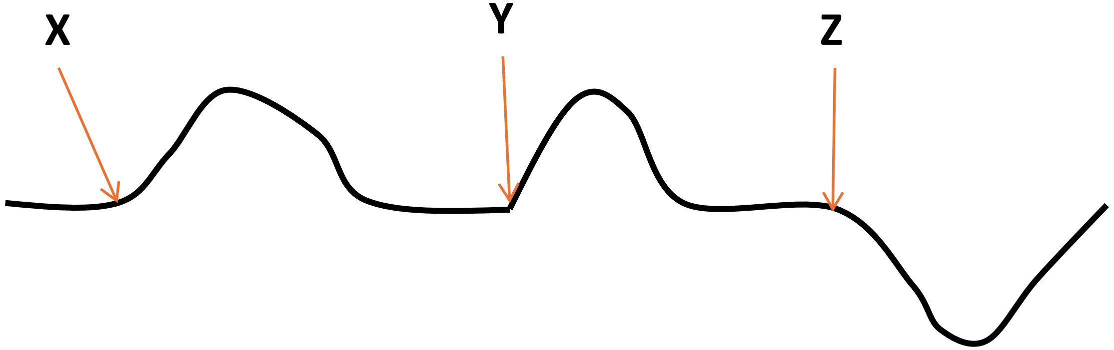
Blood Pressure

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A 95-year-old man is admitted to the intensive care unit with a diagnosis of septic shock secondary to urinary tract infection. Antibiotic therapy is initiated. Despite aggressive intravenous hydration his blood pressure is 60/40 mm Hg. An intravenous infusion of norepinephrine is initiated. Which of the following cellular changes will most likely occur directly in response to this therapy?

A. cAMP increase in vascular smooth muscle cells
B. DAG decrease in vascular smooth muscle cells
C. IP3 increase in vascular smooth muscle cells
D. cAMP increase in bronchial smooth muscle cells
E. IP3 increase in cardiac muscle cells

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SOM.MK.1.BPM1.1.FTM.4.PHAR.0052
Describe the pharmacological effects, uses and adverse effects of the endogenous catecholamines.

# FTM IMCQ 9

6 GNET Questions

Multifactorial and Karyotypes

A 35-year-old man and his wife  comes to the physician for genetic counseling because they plan to start a family. They are concerned because his brother and maternal aunt have been diagnosed with epilepsy. The couple does not have any major medical illnesses. Which of the following best describes an aspect of the disorder in the patient's family?

A. Influenced by environmental factors only
B. Transmitted as a single gene disorder
C. Risk depends on the sex of the affected individual
D. Risk factor modified by prenatal folate supplementation for mother
E. Presence of the disorder in multiple family members increases risk

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SOM.MKIII.BPM1.1.FTM.3.GNET.0116; Discuss general concepts of multifactorial disorders

A researcher studying human traits focus on those traits that can be measured. Recordings are taken regarding height, weight, blood pressure, insulin tolerance, and intelligence from 20,000 individuals and correlated to genomic data. Which of the following best describes the type of heritable traits that this researcher is studying?

A. Chromosomal aberrations
B. Autosomal dominant disorders
C. Digenic disorders
D. Quantitative traits
E. Autosomal recessive disorders
F. Mitochondrial disorders
G. X-linked disorders

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SOM.MK.III.BPM1.1.FTM.3.GNET.0116 Discuss general concepts of multifactorial disorders

A 24-year-old woman goes to a genetic counsellor to discuss her relative risk of passing on a disorder to her future children. Her family history is significant for a unique disorder that does not appear to follow basic mendelian inheritance. Both males and females are affected, however the children of affected females exhibit more clinically severe features of the disorder than children of affected males. Which of the following disorders show a similar presentation?

A. Sickle cell disease
B. Duchenne muscular dystrophy
C. Huntington’s disease
D. Pyloric stenosis
E. Familial hypercholesterolemia
F. MELAS

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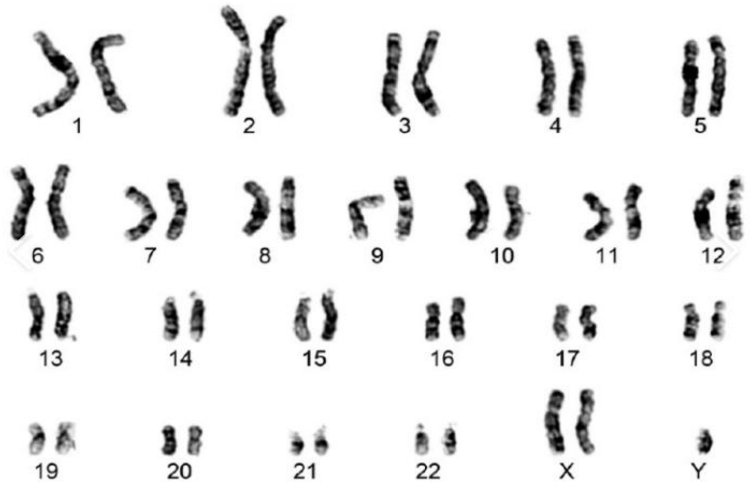
SOM.MKIII.BPM1.1.FTM.3.GNET.0117 Analyze the threshold model of multifactorial inheritance
SOM.MKIII.BPM1.1.FTM.3.GNET.0118 Analyze heritability of multifactorial disorders

A 30-year-old man comes to the physician with his wife because of a 1-year history of inability to conceive. There is no family history of major medical illness. He is 198 cm (6 ft 6 in), weighs 92.9 kg (205 lb); BMI is 23.7 kg/m$^2$ (18.5-24.9 kg/m$^2$). Physical examination shows an uncircumcised penis with testicles 2.5 cm by 2.5 cm symmetrically round, firm to palpation, and descended into the scrotum bilaterally. Laboratory studies are shown in the table. The results of his karyotype analysis is shown. Which of the following genetic mechanisms best explains the patient' disorder?

| | Patient Value | Control Value |
|---|---|---|
| Luteinizing hormone | 13 IU/L | 1.24-7.8 IU/L |
| Follicle stimulation hormone | 20 IU/L | 1.42-15.4 IU/L |
| Testosterone | 105 ng/dL | 300-1000 ng/dL |




A. Mitotic nondisjunction during embryogenesis
B. Meiosis I nondisjunction in dad
C. Reciprocal translocation in oocyte
D. Meiosis II nondisjunction in dad
E. X isochromosome formation

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SOM.MKIII.BPM1.1.FTM.3.GNET.0121: Analyze the abnormal human karyotype and define the terms euploidy and aneuploidy. Use disorders and syndromes such as Down, Patau, Edward, Turner, Klinefelter as examples

A 34-year-old woman comes to her physician for follow-up after a spontaneous loss of pregnancy at 34 weeks of gestation. Perinatal autopsy showed cleft lip and palate, bilateral polydactyly of the hands and feet, and complex congenital heart defects. Genetic testing of the fetal tissue showed a chromosomal abnormality. Which of the following best describes the fetal abnormality?

A. Trisomy 21
B. Trisomy 18
C. Trisomy 13
D. Turner syndrome
E. Kleinfelter syndrome

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SOM.MKIII.BPM1.1.FTM.3.GNET.0121
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A 2-year-old boy is brought to the physician for delays in speech and motor milestones. Physical examination shows subtle upslanting palpebral fissures and single palmar crease on both hands, but no significant cardiac or gastrointestinal anomalies. His growth parameters are within normal limits. Karyotype analysis reveals two cell lines: one with a normal 46,XY complement and another with 47,XY,+21. Which of the following genetic mechanisms best explains the clinical features seen in this patient?

A. Robertsonian translocation involving chromosome 21
B. Meiotic nondisjunction resulting in trisomy 21
C. Uniparental disomy of chromosome 21
D. Mosaicism for trisomy 21
E. Partial deletion of chromosome 21

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SOM.MKIII.BPM1.1.FTM.3.GNET.0123 Discuss mechanisms that might lead to mosiacism for chromosomal aneuploid conditions

A. Robertsonian translocation involving chromosome 21
B. Meiotic nondisjunction resulting in trisomy 21
C. Uniparental disomy of chromosome 21
D. Mosaicism for trisomy 21
E. Partial deletion of chromosome 21

# FTM IMCQ9

Biochemistry 6 Questions

A 41-year-old man is brought to the physician because of a 2-day history of pain in the left knee. He says he has difficulty sleeping at night due to severe, 8/10 pain around his left knee. He is unable to walk without assistance, and the pain persists throughout the day despite taking conventional nonsteroidal anti-inflammatory drugs. He had a similar episode 1 month ago in his right great toe. He is 180 cm (5 ft 11 in) tall and weighs 108 kg (239 lb). His temperature is 37.1 $^{o}$C (98.8$^{o}$F), pulse is 79/min, blood pressure is 136/78 mm Hg, and respirations are 17/min. His left knee is hot, erythematous, swollen, and tender. A drug that inhibits the degradation of purine nucleotides is prescribed. Which of the following enzymes is most likely targeted by this medication?

A. Adenosine deaminase
B. Ribonucleotide reductase
C. Thymidylate synthetase
D. IMP dehydrogenase
E. Xanthine oxidase
F. PRPP synthetase
G. Purine Nucleoside Phosphorylase

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0171 Correlate the basis of clinical features and explain basis of use of allopurinol and febuxostat in the management of hyperuricemia

A 39-year-old woman is brought to the emergency department because of a 1-day history of fever and diarrhea. She was diagnosed with ulcerative colitis (UC) and was prescribed 6-mercaptopurine. Her temperature is 39.4°C (102.9°F), pulse is 111/min, and blood pressure is 98/66 mm Hg. Laboratory studies are shown. A CT scan of the abdomen shows mild hepatosplenomegaly. Which of the following best explains the clinical findings in this patient?

| | **Patient** | **Reference range** |
|---|---|---|
| White blood cell (WBC) count | 14,000/mm$^3$ | 4500-11,000/mm$^3$ |
| Hemoglobin | 8.8 g/dL | 12.0-16.0 g/dL |
| Platelet count | 64,000/mm$^3$ | 150,000-400,000/mm$^3$ |
| Serum ALT | 90 U/L | 10-40 U/L |
| Serum AST | 56 U/L | 12-38 U/L |

A. Gain of function mutation in the TPMT gene
B. Homozygous recessive for a mutation in the TPMT gene
C. Increased activity in PRPP synthetase
D. Loss of function of PRPP synthetase
E. Heterozygosity for a mutation in the APRT gene
F. Loss of function mutation in the ADA gene
G. Homozygosity for a mutation in gene for xanthine oxidase

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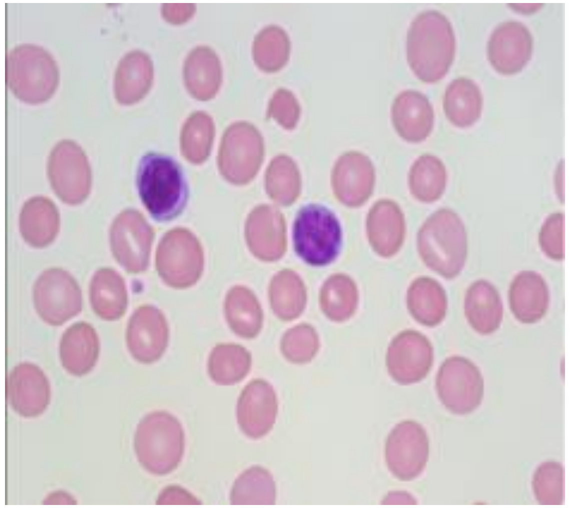
D. Loss of function of PRPP synthetase
E. Heterozygosity for a mutation in the APRT gene
F. Loss of function mutation in the ADA gene
G. Homozygosity for a mutation in gene for xanthine oxidase

SOM.MK.III.BPM1.1.FTM.3.GNET.0104; Explain the mechanism of 6-mercaptopurine/azathioprine toxicity in patients who are homozygous for autosomal recessive polymorphisms in the TPMT gene (example of an adverse drug reaction).

A 4-year-old boy is brought to the physician by his mother because of a 3-day history of bleeding from the lips. She says multiple lesions were seen on his lower lip, cheeks, and tongue. At three months of age, he showed signs of delayed psychomotor development, and at ages 1 and 4, he has had recurrent urinary tract infections. Physical examination shows the findings in the photograph. Laboratory studies show hemoglobin is 9.8 mg/dL (13-17.5) and serum uric acid level is 9.2 mg/dL (3.5-7.2). A photomicrograph of a peripheral blood smear is shown. The increased degradation of which of the following nucleobases is most likely contributing to the abnormal findings in this patient?



A. Uracil
B. Guanine
C. Adenine
D. Thymine
E. Cytosine

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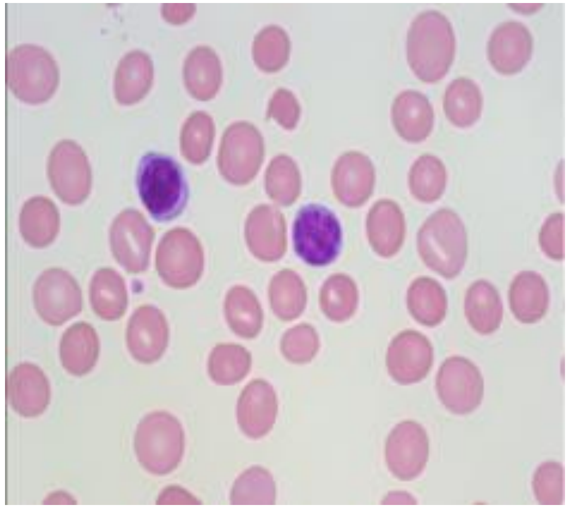

A. Uracil

✓ B. Guanine

C. Adenine

D. Thymine

E. Cytosine

SOM.MK.I.BPM1.1.FTM.3.BCHM.0168 Correlate clinical, laboratory (hyperuricemia) and molecular findings in Lesch-Nyhan syndrome (Specify enzyme deficient).

A 34-year-old man comes to the physician because of a 3-month history of intermittent abdominal pain and diarrhea. He describes the pain as crampy, localized to the right lower quadrant, and worsens after meals. Colonoscopy shows patchy, transmural inflammation with areas of ulceration and a fistulous tract. He is diagnosed with Crohn’s disease and is prescribed methotrexate as part of an induction therapy. Which of the following best describes the mechanism of action of this medication?

A. Allosteric activator of dihydrofolate reductase
B. Suicide inhibitor of thymidylate synthase
C. Competitive inhibitor of dihydrofolate reductase
D. Competitive inhibitor of IMP dehydrogenase
E. Analog of para-aminobenzoic acid
F. Suicide inhibitor of xanthine oxidase

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0165; Correlate mechanism of action of sulfa drugs, trimethoprim, methotrexate, mycophenolic acid to their clinical application

A 46-year-old woman comes to the physician because of a 2-day history of fever and loss of appetite. She underwent an orthotopic liver transplantation for end-stage liver disease seven weeks ago. She is on an immunosuppressive regimen that includes steroids and a pharmacologic agent that directly inhibits the synthesis of guanosine monophosphate. Physical examination shows right upper quadrant tenderness and hepatomegaly. Her leucocyte count is 3700/$mm^3$ (Normal: 4500-11,000/$mm^3$), alanine aminotransferase (ALT) is 90 U/L (Normal: 10-40 U/L), and total bilirubin is 1.5 (Normal: 0.1-1.0 mg/dL). Liver biopsy specimen shows an inflammatory cellular infiltrate and active CMV replication by immunofluorescence. Which of the following pharmacotherapies was most likely prescribed to the patient?

A. Hydroxyurea
B. Allopurinol
C. Trimethoprim
D. 5-fluorouracil
E. Azathioprine
F. Mycophenolic acid
G. Febuxostat

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A 4-year-old girl is brought to the hospital by her parents because of a 3-week history of low energy and pallor. Physical examination shows conjunctival pallor with poorly formed fingernails and sparse hair. Laboratory studies show macrocytic, megaloblastic anemia. Urinalysis shows orotic acid. Her anemia does not improve with B12, iron, B6, or folic acid treatment. Synthesis of which of the following is most likely impaired in this patient?

A. Adenosine and guanosine nucleotides
B. dADP and dGDP
C. Ribose 5-phosphate and PRPP
D. Hypoxanthine and xanthine
E. Uridine and thymidine nucleotides

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0178; Specify enzyme deficiency and explain metabolic basis of clinical features in orotic aciduria